Label-Free, Bias-Free Classification of Difficult 4D-STEM Diffraction with a Diffraction-Tailored Self-Supervised Model

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Abstract

Four-dimensional scanning transmission electron microscopy (4D-STEM) of beam-sensitive, low-dose, poorly-ordered materials produces diffraction data that are hard to segment into meaningful regions without labels, a forward model, or subjective choices such as the number of clusters or components. We introduce a self-supervised classifier purpose-built for nanobeam electron diffraction (NBED): a truncated convolutional DINO encoder trained with azimuthal (θ-roll) augmentation for in-plane-rotation invariance, a radial-gated supervised-contrastive objective, and a one-dimensional radial clustering loss, with emergent cluster count and optional pairwise semi-supervision. The model needs no labels, no preset K, and no per-feature thresholds. On a sodium-phyllosilicate (NaPHI) flake it reproduces our previously published, expert-derived domain segmentation label-free and without the slow SAM + orientation-mapping pipeline (Dice 0.81, IoU 0.68; $\sim$75% of the indexed domain recovered), and transfers to a new acquisition in milliseconds. A built-in interpretability protocol (representational probing, causal ablation, classical-baseline comparison, and ACOM cross-check) shows the clusters are governed by scattered intensity and low-q two-dimensional diffraction structure, are rotation-invariant, and quantifies when the model is distinctive versus classical analysis: highly distinctive on hard low-dose molecular films (indomethacin, IMC) where classical decompositions cannot produce a coherent map, and convergent with classical analysis on well-ordered samples (NaPHI, EuInAs). The method is most valuable exactly where existing tools fail.

Keywords

4D-STEM, nanobeam electron diffraction, self-supervised learning, unsupervised clustering, low-dose, interpretability, indomethacin

Introduction

4D-STEM records a full diffraction pattern at every probe position, encoding local structure, orientation, strain and disorder across a field of view.[1] Turning that data cube into a meaningful map of distinct regions is a prerequisite for almost any downstream analysis. For robust, well-ordered crystals this is routine; for the beam-sensitive, low-dose, partially-ordered materials of greatest current interest—organic semiconductors, pharmaceutical polymorphs, layered and defective crystals—it is not. Patterns are sparse and noisy, domains are numerous, and diffuse scattering blurs the distinctions a human or a template would key on.

Unsupervised and self-supervised analysis is attractive because it needs no labels.[2,3] But the standard automated routes each demand a subjective choice the microscopist must defend: k-means and density clustering require the number of clusters (or a density scale); non-negative matrix factorization (NMF) and principal component analysis (PCA) require the number of components;[4] supervised segmentation requires labels, which on hard data is prohibitive; and segmentation-model pipelines (e.g. SAM) plus orientation mapping are slow and parameterized. Just as importantly, even when a clustering succeeds, its *physical* meaning is rarely established—so the map cannot be trusted.

Here we address both problems. We present a self-supervised classifier engineered specifically for NBED, which produces a label-free, threshold-free first segmentation; we validate it against an independent, previously published expert result; and we pair it with an interpretability protocol that establishes what the clusters physically represent and when the method adds value over classical analysis.

Results and Discussion

A diffraction-tailored self-supervised model

Our classifier is a self-distillation (DINO) network,[2] but every major design choice is dictated by the physics of NBED rather than inherited from natural-image vision:

- **Convolutional, truncated encoder.** We use a truncated ResNet rather than a vision transformer. NBED patterns are small, sparse and low-count; the convolutional inductive bias and the reduced depth match the available signal, whereas transformers are data-hungry and overparameterized for this regime.
- **Physics-chosen augmentations.** An azimuthal "θ-roll" augmentation rotates the pattern in-plane during training, enforcing invariance to in-plane crystal rotation—a diffraction pattern rotated about the optic axis represents the *same* structure. Flips and colour-jitter, standard in vision, are disabled because they are meaningless or harmful for diffraction.
- **Radial-gated contrastive + 1-D loss.** A radial-gated supervised-contrastive term[5] and a one-dimensional radial clustering loss bias grouping toward the physically meaningful radial/ring signature.
- **Emergent K and optional semi-supervision.** We over-specify the number of prototypes; the model populates only those it needs, so the active cluster count emerges from the data rather than being preset. A small number of pairwise "same/different" hints can optionally steer the partition without reverting to full labelling.

The result is a single forward pass from raw cube to a class map, with no user-set K, components, or thresholds.

Label-free concordance with expert segmentation

We first establish trust on a sample with an independent, published answer. In prior work on a NaPHI flake we segmented an oriented "Line" domain using a SAM segmentation plus orientation mapping with user-defined line parameters.[6] Running the present model on the same scan, with no labels and no line parameters, the union of two emergent classes reproduces that domain (Figure **??**): Dice = 0.81, IoU = 0.68, with ∼75% of the published domain recovered and ∼89% of the model's domain falling within the published one. The model achieves this without the SAM step, and a model trained on this flake transfers to a second, tilted acquisition in milliseconds—turning a slow, parameterized pipeline into an automatic one.

Resolving low-dose molecular films where classical methods fail

The payoff is on the hard target. On low-dose indomethacin (IMC) films the model produces coherent, fine-grained class maps, whereas classical automated methods do not. Clustering virtual dark-field, azimuthal-profile, PCA

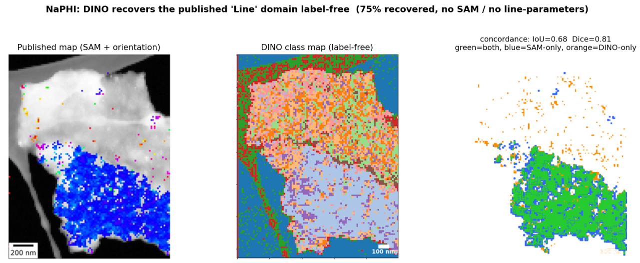

Figure 1. NaPHI: the label-free DINO class map (centre) versus our previously published SAM+orientation segmentation (left), same scan. Overlay (right): green = agreement. The model recovers the oriented domain (Dice 0.81, IoU 0.68) with no labels, no SAM and no line parameters.

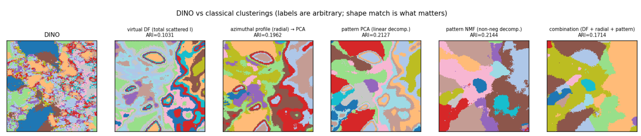

Figure 2. IMC (low-dose molecular): the DINO map (far left) versus classical clusterings of the same data (virtual-DF, azimuthal, PCA, NMF, combination). Colours are arbitrary; spatial shape is what matters. Classical methods recover only coarse regions (best ARI ≈ 0.15).

and NMF features of the *same* data into the same number of groups agrees only weakly with the DINO map (best adjusted Rand index, ARI $\approx$ 0.11–0.21 across three IMC films; Figure **??**). There is no external ground truth here—classical analysis simply cannot deliver a usable map—so the method is the only practical route, and its trustworthiness rests on the validation above and the interpretability below.

What the clusters encode

To establish meaning we apply four complementary tests (Figure **??**). *Representational probing*—5-fold ridge regression from the frozen embedding to candidate physical factors, with the correlation ratio η^2—shows the embedding most strongly encodes the total scattered intensity and crystallinity, and essentially ignores crystallographic orientation and Bragg-peak count. *Causal ablations* (re-inference on perturbed inputs, scored by ARI against the original map) disentangle the contributions. A consistent post-mask intensity normalization (rescaling the kept, beam-masked region to constant) collapses the intensity-dominated samples (NaPHI, EuInAs → 1–2 classes) but leaves IMC with structure (7 classes, ARI 0.12): IMC depends on scattered intensity *and* low-q 2-D structure, not intensity alone. A complementary *per-ring* normalization—flattening the radial profile while preserving within-ring angular structure—collapses the molecular samples (NaPHI, IMC; i.e. radial/intensity-driven and rotation-invariant) but EuInAs retains ARI 0.29, showing it additionally uses angular/orientation structure, consistent with its zone-axis correlation (AMI ≈ 0.30). Azimuthal averaging collapses all samples (the full 2-D pattern is needed, not a 1-D profile), and masking the low-q band is far more damaging than high-q. *Classical-baseline comparison* quantifies novelty (above). *ACOM cross-check* confirms the partition is uncorrelated with indexed zone axis on the molecular samples (adjusted mutual information ≈ 0), consistent with the θ-roll-imposed rotation invariance. Together: the clusters are driven by scattered intensity and low-q 2-D diffraction structure, are rotation-invariant, and their distinctiveness from classical analysis is material-dependent.

Generality: crystalline, multi-phase, oriented data

The approach is not limited to molecular films. On a layered EuInAs sample with multiple phases, the model produces a sensible multi-domain map; classical methods here recover about half of it (ARI ≈ 0.44), i.e. the model is *least* distinctive on well-ordered material—it is most distinctive precisely on the hard molecular case. A three-phase ACOM analysis (EuInAs/In/As) further shows that, for this crystalline sample, the DINO classes additionally track crystal orientation (zone-axis AMI ≈ 0.30), a genuine secondary axis that the molecular films lack.

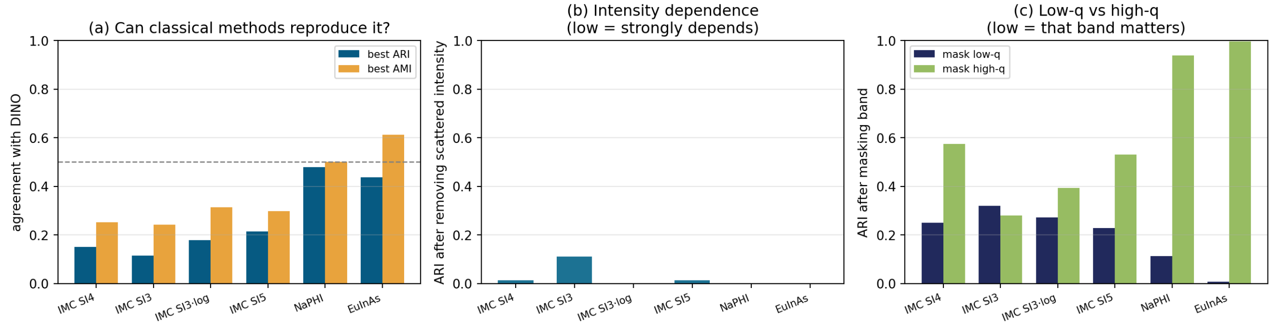

Figure 3. Cross-sample interpretability (IMC SI3/SI4/SI5, NaPHI, EuInAs). (a) classical agreement with the DINO map (rises from distinctive IMC to substantially-captured NaPHI/EuInAs); (b) intensity dependence (low bar = strongly depends); (c) low-q vs high-q (low bar = that band matters—NaPHI/EuInAs depend almost entirely on low-q).

A graphical interface for preparation, analysis and comparison

To make the method usable by experimentalists who do not program, we provide a graphical application built around three roles.

(1) Data preparation. It loads the common cube formats (.prz/.npy/ .npz/.h5, including detector-linked masters), and offers the preprocessing that matters for hard data: intensity scaling (linear $v_{\max}$ or logarithmic stretch), centre masking, real- or reciprocal-space Gaussian binning for low-dose denoising, region-of-interest selection, and multi-dataset handling —all non-destructive, writing new derived datasets.

(2) Output analysis. This is the core value. The class map is fully interactive: clicking a position shows its raw pattern; clicking a class shows the *class-average* diffraction; grains can be averaged individually; and single-frame, per-grain and per-cluster averages can be stacked and compared side by side, with on-image q/d-spacing readout and gradient-attribution overlays. A one-click "Interpretation" tab runs the full battery above (probing, ablation, classical-baseline, ACOM cross-check) and writes a report —so the question "what do the clusters mean?" is answered inside the same tool that produced them.

(3) Classical analysis, for comparison or standalone. The same interface runs the classical methods on the same map—automated crystal orientation mapping (ACOM), NMF + clustering, crystallinity and spottiness mapping, and SAM-based segmentation—so a user can directly compare the self-supervised map against any of them, or use the classical tools on their own. The model is thus offered not as a black-box replacement but as the fast, unbiased first pass within a complete, comparison-ready toolkit.

Conclusions

We have introduced a self-supervised, diffraction-tailored classifier that turns difficult low-dose 4D-STEM into a label-free, threshold-free, and physically interpretable domain map. It reproduces an independent published expert segmentation without labels or SAM, transfers across acquisitions in milliseconds, and—uniquely among the methods we compared—extracts a coherent map from low-dose molecular films that classical decompositions cannot. A built-in interpretability protocol establishes that the clusters encode scattered intensity and low-q diffraction structure rather than orientation, and quantifies where the method adds value. Because it removes the subjective choices (labels, K, components, thresholds) that gate existing workflows while remaining auditable, it is well suited as a first-pass, high-throughput tool for beam-sensitive crystalline materials.

Methods

Model and training. Truncated ResNet encoder; DINO self-distillation with teacher/student temperatures and EMA; azimuthal θ-roll augmentation (student/teacher ranges), flips and colour-jitter disabled; radial-gated supervised-contrastive term and 1-D radial clustering loss; over-specified prototype count with emergent active K; optional pairwise semi-supervision. **Data.** NBED cubes of IMC (low-dose molecular), NaPHI and EuInAs. **Interpretability.** Ridge-probe R^2 and η^2; causal input ablations re-inferred and scored by ARI; classical baselines (virtual-DF, azimuthal-PCA, pattern-PCA, NMF) clustered to the active K and scored by ARI/AMI; ACOM via py4DSTEM.[7,8] **Concordance.** IoU/Dice/containment of aligned, same-grid class and reference maps.

Acknowledgement Analyses were performed on frozen, previously validated model checkpoints.

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